Insight-V: Exploring Long-Chain Visual Reasoning with Multimodal Large Language Models

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Abstract

Multimodal large language models (MLLMs) have made rapid progress, yet they still struggle with long-chain, step-by-step visual reasoning due to limited high-quality supervision and training paradigms that entangle reasoning with answering. We introduce Insight-V, a framework that separates visual reasoning from summarization and couples it with scalable long-chain data generation and iterative preference optimization. A progressive generator produces JSON-structured chains with action signals, while multi-granularity assessment filters incorrect paths and scores candidates to select high-quality traces. We train a reasoning agent with supervised fine-tuning (SFT) followed by iterative Direct Preference Optimization, and a summary agent that learns to use or ignore reasoning traces to preserve perception. Applied to LLaVA-NeXT-8B and a 7B MLLM, Insight-V yields consistent gains on diverse reasoning benchmarks while maintaining or improving general perception. For example, the average across seven benchmarks improves from 40.2 to 47.2 (+7.0) on LLaVA-NeXT-8B, and from 48.7 to 51.6 (+2.9) on a 7B base, with increases on TextVQA, DocVQA, OCRBench, and AI2D.

1. Introduction

Large multimodal models (LMMs) built by coupling a vision encoder with a language model have achieved impressive general-purpose capabilities through visual instruction tuning [1, 9, 11, 20]. Despite this progress, these models often underperform on tasks that require long-chain, compositional visual reasoning. Two factors contribute to this gap: (1) a lack of scalable, high-quality supervision for multi-step visual chains of thought, and (2) training recipes that entangle reasoning and answer production, leading to error propagation and degraded perception.

We propose Insight-V, a framework that disentangles reasoning from summarization and trains dedicated agents for each role on long, structured chains. Our key idea is to generate diverse, JSON-formatted reasoning traces with explicit actions (continue/summary), assess them with both language and multimodal judges, and then train (i) a reasoning agent via SFT and iterative preference optimization and (ii) a summary agent that learns when to use or ignore provided thoughts. This separation improves robustness to flawed chains and stabilizes training.

Our contributions are: (i) a scalable pipeline for long, structured visual reasoning traces with progressive generation and multi-granularity assessment; (ii) a multi-agent MLLM training recipe that disentangles reasoning and summarization; (iii) an iterative Direct Preference Optimization (DPO) procedure [22] that strengthens long-chain stability using model-generated preferences, complementing RLHF [2]; and (iv) consistent gains across diverse visual reasoning and perception benchmarks, demonstrating generalizability across backbones.

2. Method

Insight-V decouples chain-of-thought generation from answer production via two agents that share a vision encoder but specialize their heads. The reasoning agent emits JSON traces with ordered actions and confidences. We sample progressively, expand partial chains, verify steps with a light checker, rescore with an assessor, and select top-k diverse traces. The reasoning agent is first SFT-trained on traces, then refined with iterative Direct Preference Optimization from assessor- and task-derived preferences. The summary agent conditions on image, question, and an optional trace; it learns to compress faithful traces or ignore low-confidence ones to preserve perception and avoid overthinking. Unless noted, both agents use the same tokenizer and backbone; only instruction adapters and losses differ.

3. Related Work

Multimodal LLMs and instruction tuning. LLaVA and follow-ups demonstrate strong general-purpose visual chat capabilities by connecting CLIP-like encoders to LLM backbones and instruction tuning on curated data [9, 11]. Qwen-VL/Qwen2-VL extend versatile perception and localization abilities [1, 20]. Our framework builds on these foundations but focuses on long-chain reasoning stability rather than only single-turn instruction following.

Preference learning and alignment. RLHF [2] im-

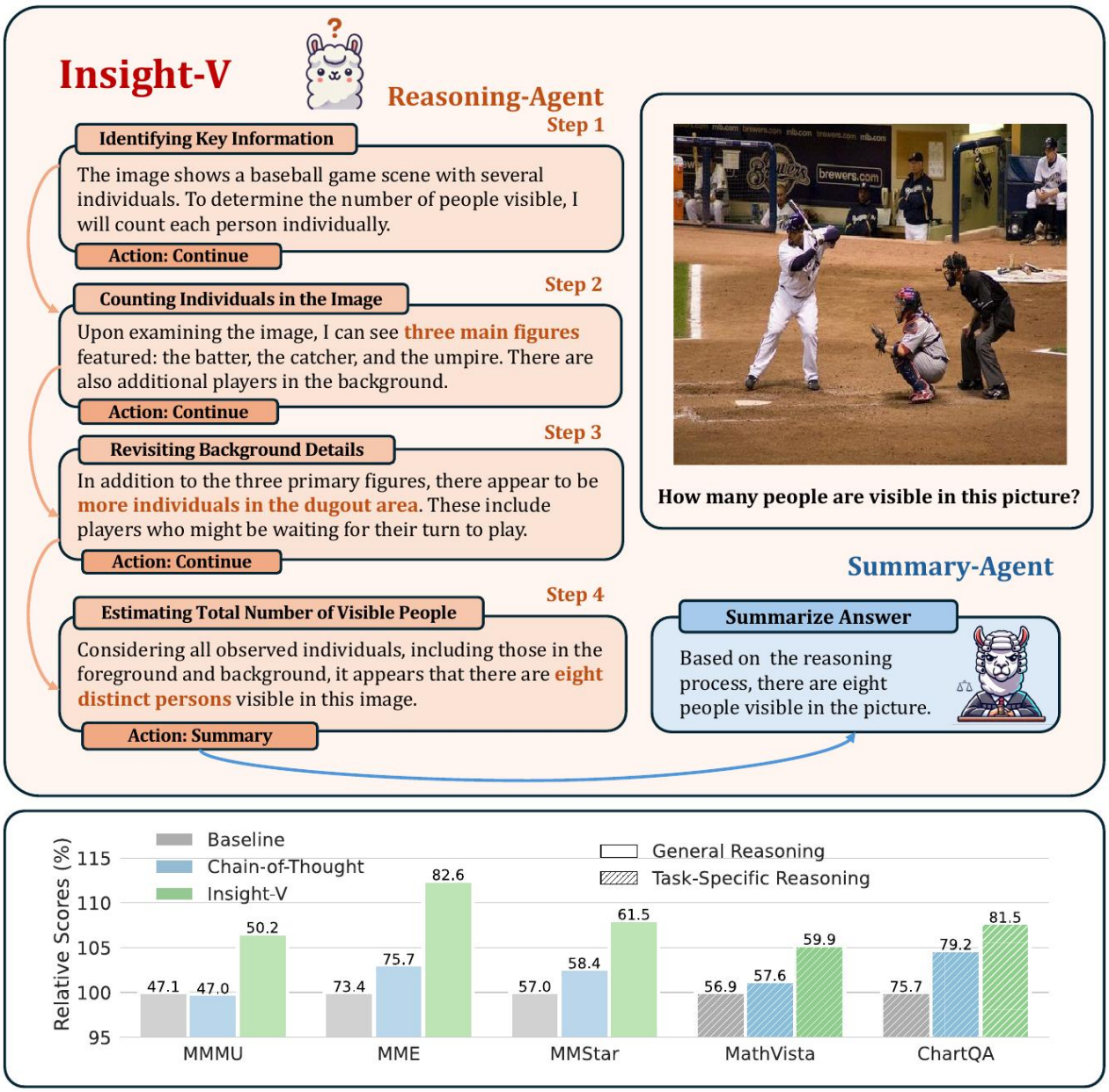

Figure 1. Insight-V at a glance: a reasoning agent explores long-chain visual thoughts and a summary agent produces concise answers. Gains are consistent across diverse benchmarks.

proves helpfulness and safety but can be unstable and expensive. DPO [22] simplifies preference optimization via a classification-style objective. We adopt an *iterative* DPO procedure over model-generated chain pairs to enhance long-chain stability.

Benchmarks for visual reasoning and perception. We evaluate on MMMU and MMMU-Pro [28, 29], MMBench [12], MathVista [16], ChartQA [18], MMStar [3], and perception-oriented datasets including TextVQA [23], DocVQA [19], OCRBench [13], and AI2D [6]. We reference CLIP [21] as the vision backbone commonly used in these systems.

Reasoning traces vs. direct answers. Prior instruction tuning commonly supervises final answers or short rationales, intertwining explanation tokens with answers and risking perception drift if noisy steps are imitated [1, 9]. Our approach separates a long-horizon reasoning policy from a summary policy and trains them on distinct signals (step

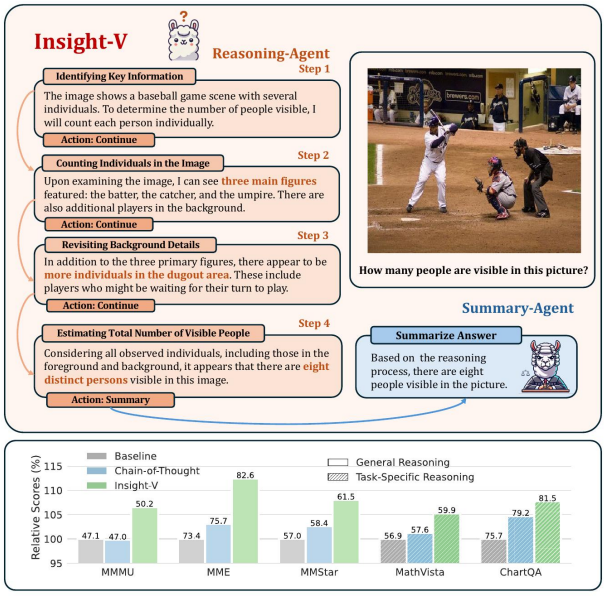

Figure 2. Illustration and performance of Insight-V.

Algorithm 1 Insight-V training with iterative DPO

Initialize reasoner θ_r and summarizer θ_s
for round $= 1 \ldots R$ **do**
  Generate budgeted chains via progressive sampling
  Assess steps; select top-k diverse traces
  SFT update of reasoner on selected traces
  Build (chosen, rejected) pairs at traj/window levels
  DPO update of θ_r on preference pairs
  Freeze θ_r; train θ_s to summarize or skip
end for
Return θ_r, θ_s

trajectories versus short answers). This decoupling reduces error propagation from imperfect chains and improves robustness while preserving general perception.

4. Methodology

Insight-V comprises (i) a scalable data pipeline for long-chain reasoning traces, (ii) a two-agent training paradigm that separates reasoning and summarization, and (iii) an iterative preference optimization stage for stability. Figure 3 overviews the model design and Figure 4 outlines data generation.

Progressive long-chain generation. We employ a progressive generator that emits JSON-formatted steps with fields such as `title`, `content`, and `next_action` (`continue` or `final_answer`). The generator conditions on the image and prompt, producing at least two steps and exploring alternative hypotheses with sampling (e.g., temperature 0.7, top-p 0.95) to induce diverse paths.

Multi-granularity assessment. We first filter wrong answers using a strong text LLM to verify short-form answers when available, then score surviving chains with a capable MLLM judge (e.g., Qwen2-VL) considering consistency, visual grounding, and step quality. We retain top-ranked chains per query to form high-quality supervision.

Reasoning agent. From the top chains ($\tilde{2}$00K examples), we train a reasoning agent via SFT to imitate high-quality step trajectories. We then refine the agent with *iterative* DPO: over three rounds, we sample preferred and dispreferred chains ($\tilde{1}$5K pairs/round; learning rate $5e^{-7}$) and update the policy, re-generating candidates for the next round.

Summary agent. We train a separate summary agent to produce concise, accurate answers given the image, prompt, and optionally the reasoning enclosed in custom tags (e.g., `<thoughts>...</thoughts>`). The training set mixes optimal and flawed chains and general QA ($\tilde{1}$.2M examples), enabling the agent to robustly *use or ignore* provided reasoning to preserve perception and avoid propagating errors.

Backbones and connectors. We instantiate Insight-V with LLaVA-NeXT-8B (CLIP ViT-L/14 @336px and a 2-layer MLP connector) [9, 10, 21] and a 7B MLLM based on the Qwen family [1, 20]. For Qwen-based models, we leverage pretraining on captions and SFT data as provided in the base model release.

Assessment and filtering. For each progressive sample we apply step-level and trajectory-level checks using two complementary judges: a cross-modal scorer that verifies image–text consistency of referenced regions, quantities, and relations, and a language-only verifier that flags circular reasoning or contradictions. We enforce structural constraints (valid JSON schema, required action tokens, bounded depth) and canonicalize units and coordinates. Candidate traces are ranked by a weighted mix of coverage, correctness proxies, and brevity; we retain top-k and synthesize hard negatives via step perturbations. These positives/negatives seed DPO preferences, reducing noise while preserving diverse reasoning styles.

Training and inference. During SFT we freeze the vision encoder and connector for the first epoch, then unfreeze the connector to protect perception while adapting the LM. We cap chain length at 16 steps and add a mild length penalty during sampling to avoid loops. At inference the reasoning

agent receives a budget of up to 8 steps; if internal quality signals fall below a threshold, we elide thoughts and ask the summary agent to answer directly, which reduces error propagation on short questions.

Instruction formatting and schema. Inputs prepend an image token and a brief instruction requesting stepwise reasoning. The model emits actions via `<think>`, `<cont>`, and `<sum>`. Each step is a compact JSON object with fields `role`, `thought`, `evidence`, and `action`; evidence points to region indices or text spans. We canonicalize numbers, units, and box coordinates to stabilize training.

Optimization details. SFT minimizes next-token loss with teacher forcing over actions; tokens inside OCR spans receive lower weight to avoid fitting glyph noise. Preference learning builds (chosen, rejected) pairs at trajectory and window levels and linearly anneals the preference weight, with early stopping on a small reasoning dev set. Decoding constrains action tokens and applies a mild repetition penalty to curb loops.

Why separation helps. Jointly training a single head to both plan and answer creates gradient interference between perception tokens and abstract reasoning tokens. Decoupling converts hidden thought into an explicit latent that can be pruned, perturbed, or dropped at inference time. In practice we observe fewer degenerate loops, better calibration of stop/continue actions, and improved transfer to perception-heavy tasks because the summary policy is never forced to imitate noisy chains.

Complexity and latency. Training cost scales roughly linearly with chain length and the number of preference pairs. At inference, the budgeted reasoner adds a small, controllable overhead proportional to the number of steps; decoding remains streaming-friendly because action tokens are drawn from a tiny constrained set. Memory is dominated by the LM KV cache; our 8-step budget fits comfortably within typical 8B/7B serving limits.

5. Experiments

Setup. We evaluate on seven visual reasoning benchmarks and four perception-oriented datasets. Unless noted, evaluation follows community toolchains (e.g., multiple-choice accuracy for MMMU/MMMU-Pro/MMBench, exact match for ChartQA, and official scoring for MathVista and MMStar). MathVista is judged by GPT-4-0613 following common practice.

Protocol. We use deterministic decoding (temperature 0) and official validation splits and normalizers. For generative tasks with multiple references, we report the best exact match

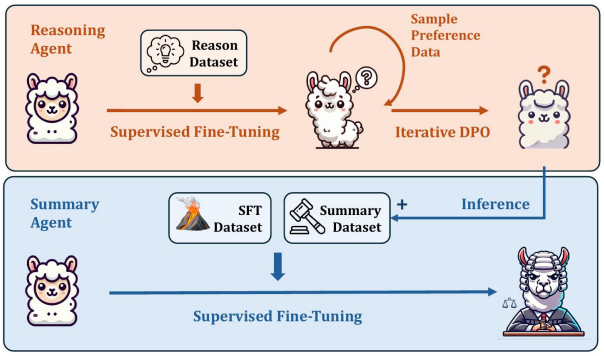

Figure 3. Overview of Insight-V model design with separate reasoning and summary agents.

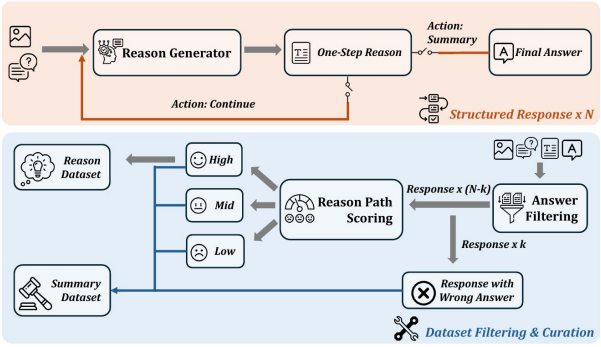

Figure 4. Data generation pipeline for long, structured reasoning traces with progressive sampling and multi-granularity assessment.

under official scripts. Baselines are reproduced with the same evaluation toolchain or taken from public leaderboards for the same backbones to ensure fair comparability.

Scoring and normalization. For multiple-choice tasks we apply official tie-breaking and length-normalized logits when supported; for ChartQA we standardize units and rounding prior to exact match. We fix seeds and disable sampling (temperature 0, top-p 1.0) for replicability. Unless a benchmark mandates cropping, inputs use the full image at the backbone's native resolution.

Main results on visual reasoning. Table 1 reports results across MMMU, MMMU-Pro, MMBench, MME, ChartQA, MMStar, and MathVista. On LLaVA-NeXT-8B, the average improves from 40.2 to 47.2 (+7.0), with notable gains on MMMU (36.9→42.0), MMMU-Pro (13.2→21.0), MMBench (72.3→81.7), ChartQA (69.4→77.4), MMStar (43.1→57.4), and MathVista (45.9→49.8). On a stronger 7B base, the average rises from 48.7 to 51.6 (+2.9), with consistent improvements across tasks.

General perception. Table 2 shows that Insight-V maintains or improves general perception: on LLaVA-NeXT-8B we observe increases on TextVQA (65.2→70.5),

Table 1. **Results on Visual Reasoning Tasks.** We conduct evaluation experiments across 7 benchmarks, covering both general reasoning and task-specific reasoning assessments. Insight-V exhibits notable effectiveness and generalizability when applied to LLaVA-NeXT and our baseline model, surpassing other state-of-the-art MLLMs by a large margin.

Model	Size	MMMU	MMMU-Pro	MMBench	MME	ChartQA	MMStar	MathVista	Average
DeepSeek-VL [15]	7B	35.4	–	73.5	–/–	59.1	37.1	36.1	–
VILA-1.5 [8]	8B	38.6	–	75.3	1634.9/–	–	39.7	–	–
Cambrian-1 [24]	8B	42.7	–	75.9	1547.1/–	73.3	–	49.0	–
InternLM-XComposer2 [4]	7B	41.1	–	77.6	2220.4	71.8	56.2	59.5	–
POINTS [14]	7B	51.4	–	78.0	2184.1	–	60.9	63.0	–
IXC-2.5 [31]	7B	42.9	–	79.4	2233.1	82.2	59.9	**63.7**	–
Bunny-LLaMA3 [5]	8B	43.4	–	77.2	1588.9/321.1	–	–	34.4	–
MM-1.5 [30]	7B	41.8	–	–	1514.9/346.4	78.6	–	47.6	–
MiniCPM-LLaMA3-V 2.5 [25]	8B	45.8	19.6	77.2	2024.6	–	51.8	54.3	–
MiniCPM-V-2.6 [26]	7B	49.8	**27.2**	78.0	2268.7	–	57.5	60.6	–
Qwen2-VL [20]	7B	**53.7**	–	81.0	–	**83.0**	60.7	61.4	–
Idefics3-LLaMA3 [7]	8B	46.6	22.9	77.5	1937.4	74.8	55.9	58.4	48.1
Ovis1.5-LLaMA3 [17]	8B	48.3	23.6	76.6	1948.5	76.4	57.3	63.0	49.4
LLaVA-NeXT-LLaMA3 [10]	8B	36.9	13.2	72.3	1611.1/346.0	69.4	43.1	45.9	40.2
+ Multi-Agent	8B	40.8	17.8	77.6	1603.7/469.3	74.6	52.6	47.4	44.5
+ Iterative DPO (**Insight-V-LLaVA**)	8B	42.0	21.0	81.7	1583.9/485.4	77.4	57.4	49.8	**47.2** (+7.0)
Our Base Model	7B	47.1	22.6	81.3	1573.7/482.5	75.7	57.0	56.9	48.7
+ Multi-Agent	7B	49.7	23.8	82.2	1662.2/**629.3**	81.2	58.6	58.7	50.7
+ Iterative DPO (**Insight-V**)	7B	50.2	24.9	**82.3**	**1685.1**/627.0	81.5	**61.5**	59.9	**51.6** (+2.9)

Table 2. **Results on other multimodal benchmarks.** Insight-V enhances reasoning capabilities without compromising general visual perception and even achieves improvements on benchmarks requiring perception ability more.

Model	TextVQA	DocVQA	OCRBench	AI2D
LLaVA-NeXT-LLaMA3	65.2	78.2	553	71.5
+ Multi-Agent	68.9	81.8	631	75.7
+ Iterative DPO (**Insight-V-LLaVA**)	**70.5**	**82.9**	**663**	**77.3**
Our Base Model	75.4	90.2	713	79.7
+ Multi-Agent	**77.0**	91.4	**738**	**80.1**
+ Iterative DPO (**Insight-V**)	76.8	**91.5**	735	79.8

DocVQA (78.2→82.9), OCRBench (553→663), and AI2D (71.5→77.3). The 7B base model remains strong, with small gains or parity.

Implementation notes. For LLaVA-NeXT-8B, we use CLIP ViT-L/14 @336px with a 2-layer MLP connector [9, 21]. For the 7B base, we rely on Qwen-family MLLMs with available caption pretraining and SFT data [1, 20]. Training hyperparameters follow Section 4 where applicable; hardware details are omitted per guidelines.

6. Analysis

Ablations on design choices. Table 3 studies the contribution of each component. The multi-agent design (62.1 Avg) outperforms direct SFT (60.6), multi-turn supervised training (61.0), and summary-only (59.8), highlighting the benefit of disentangling reasoning and summarization.

Ablations on preference optimization. Table 4 compares preference learning strategies. Iterative DPO achieves the best average (63.3) over single-pass DPO (62.7) and RLAIF (62.3), indicating that repeated preference refinement improves long-chain robustness [22, 27]. Figure 6 shows

Table 3. **Ablations on the Insight-V Design Choice.** The multi-agent design outperforms other configurations, highlighting the critical role of reasoning and summarization decomposition.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Baseline	47.1	75.7	56.9	57.0	59.2
Vanilla – Direct SFT	47.0	79.2	57.6	58.4	60.6
Multi-Turn Supervised	48.1	79.6	57.9	58.2	61.0
Summary Agent Only	47.5	76.3	57.3	57.9	59.8
Multi-Agent	49.7	81.2	58.7	58.6	62.1

Table 4. **Ablations on the DPO training strategy.** Iterative DPO progressively enhances the model's reasoning capabilities, leading to improved performance.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Insight-V (Multi-Agent)	49.7	81.2	58.7	58.6	62.1
+ RLAIF	49.5	81.4	59.1	59.2	62.3
+ DPO	50.8	80.8	59.3	59.9	62.7
+ Iterative DPO	50.2	81.5	59.9	61.5	63.3

performance trends versus training data volume, suggesting that both data scale and preference refinement contribute to gains.

Error analysis. Residual failures cluster into two groups: (i) fine-grained text recognition and layout reasoning in dense documents and charts, and (ii) arithmetic chains where an early slip corrupts later steps. The two-agent design helps by letting the summary agent override low-confidence chains, but perception-bound cases persist, suggesting that stronger OCR will further help.

Qualitative analysis. Figure 8 showcases examples where the reasoning agent explores alternatives and the summary agent distills accurate answers or overrides misleading thoughts, improving robustness to flawed chains. **Chain-length sensitivity.** Budgeted reasoning improves perfor-

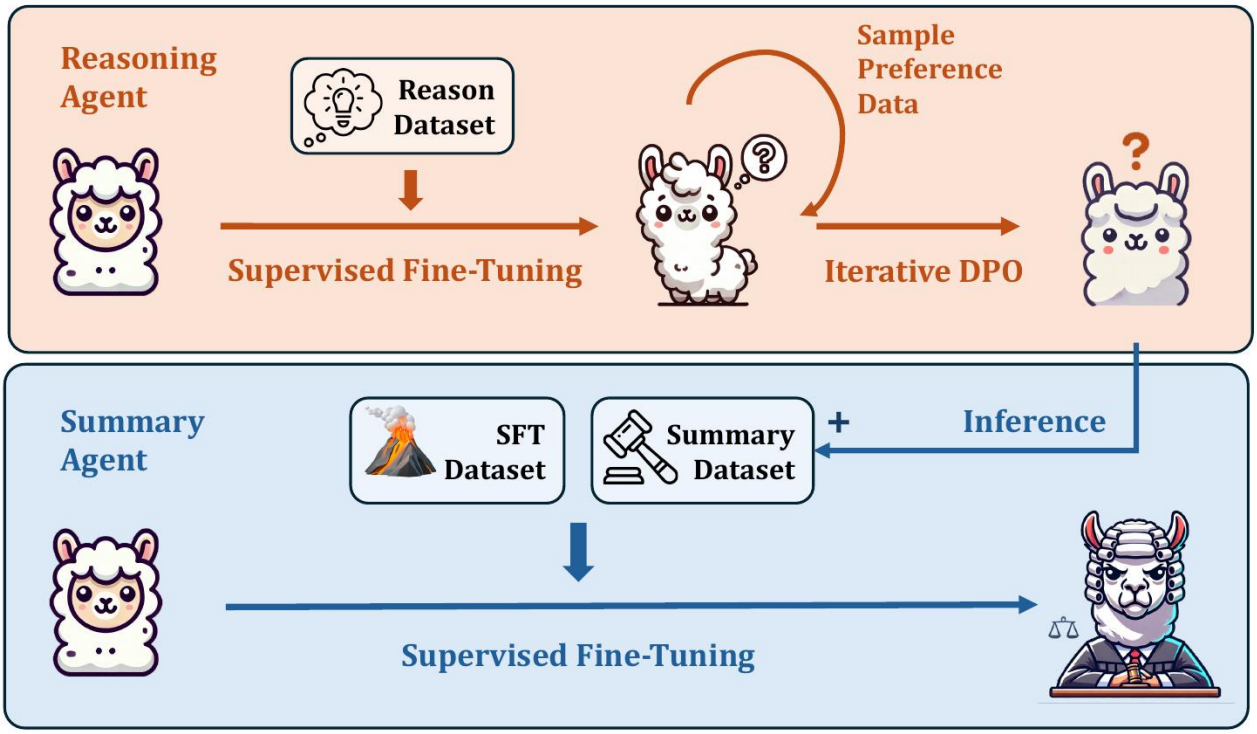

Figure 5. Method overview. Insight-V separates reasoning from summarization and combines scalable long-chain data generation with iterative preference optimization.

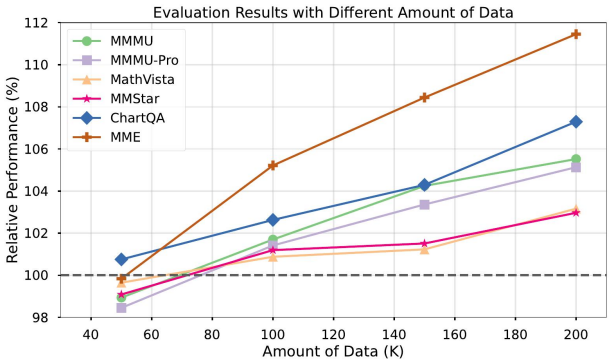

Figure 6. Ablations on the amount of training data.

mance up to a moderate depth, with diminishing returns beyond 8–12 steps. Very long chains can accumulate small locality or arithmetic slips; the summary agent mitigates this by truncating low-confidence thoughts and answering directly when appropriate.

Per-task trends. On TextVQA and DocVQA, most wins come from disambiguating fine-grained references in dense layouts. On ChartQA and MathVista, improvements concentrate on multi-hop arithmetic and unit normalization, where progressive chains help surface intermediate quantities for verification.

Answerability calibration. When the reasoning agent signals low internal quality, deferring to direct answering reduces overthinking on simple prompts and avoids hallucinated steps, yielding small but consistent gains on MMBench-style multiple choice.

7. Conclusion

We presented Insight-V, a multi-agent framework for long-chain visual reasoning that couples scalable data generation with iterative preference optimization. By separating reasoning from summarization and training each component on high-quality supervision, Insight-V delivers consistent improvements on diverse reasoning benchmarks without sacrificing general perception. Our results support the view that decoupling thought processes from answer production, followed by preference-based refinement, is an effective recipe for stabilizing and enhancing long-chain visual reasoning in MLLMs.

References

[1] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language

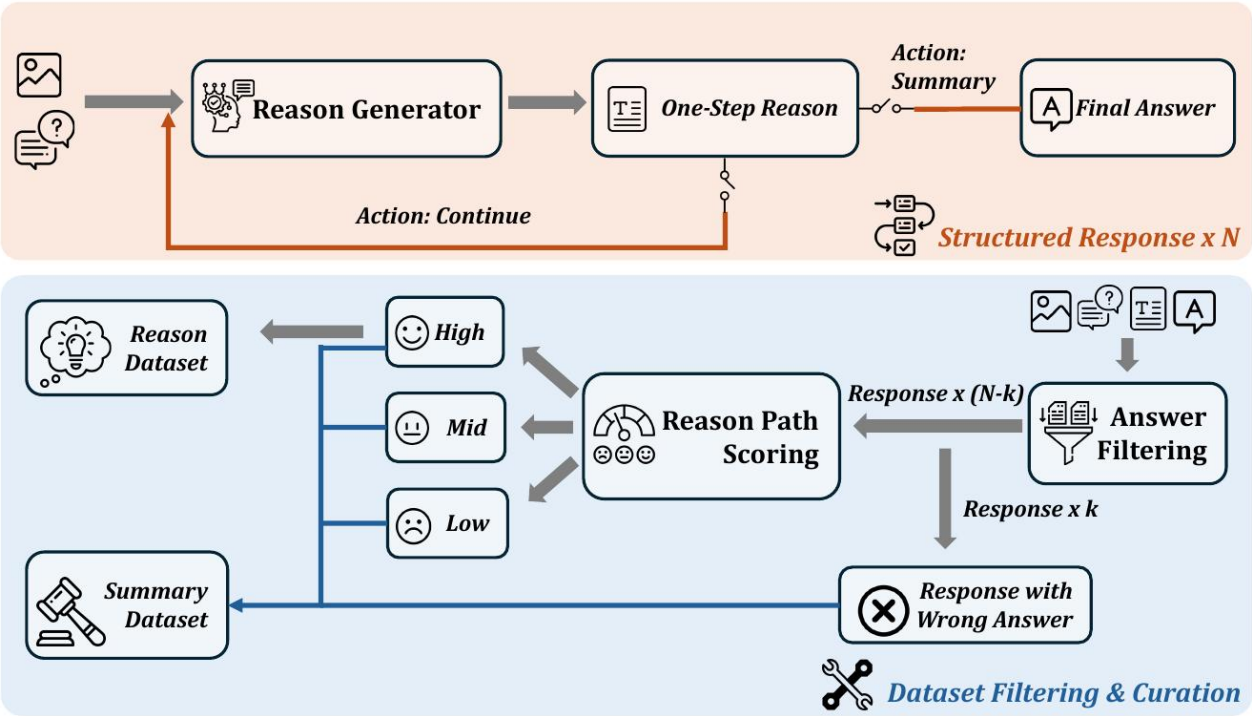

Figure 7. Data pipeline. Progressive generation, multi-granularity assessment, and preference optimization yield high-quality long-chain traces for training.

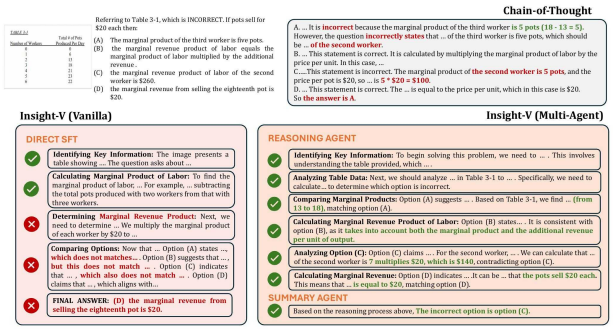

Figure 8. Qualitative results for Insight-V. Left: dense-document and chart cases where explicit chains help disambiguate references. Right: arithmetic examples where the summary agent overrides low-confidence thoughts to prevent error propagation.

model for understanding, localization, text reading, and beyond. 2023. 1, 2, 3, 5

[2] Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022. 1

[3] Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, et al. Are we on the right way for evaluating large vision-language models? *arXiv preprint arXiv:2403.20330*, 2024. 2

[4] Xiaoyi Dong, Pan Zhang, Yuhang Zang, Yuhang Cao, Bin Wang, Linke Ouyang, Xilin Wei, Songyang Zhang, Haodong Duan, Maosong Cao, et al. Internlm-xcomposer2: Mastering free-form text-image composition and comprehension in vision-language large model. *arXiv preprint arXiv:2401.16420*, 2024. 5

[5] Muyang He, Yexin Liu, Boya Wu, Jianhao Yuan, Yueze Wang, Tiejun Huang, and Bo Zhao. Efficient multimodal learning from data-centric perspective. *arXiv preprint arXiv:2402.11530*, 2024. 5

[6] Aniruddha Kembhavi, Mike Salvato, Eric Kolve, Minjoon Seo, Hannaneh Hajishirzi, and Ali Farhadi. A diagram is worth a dozen images. In *ECCV*, pages 235–251. Springer, 2016. 2

[7] Hugo Laurençon, Andrés Marafioti, Victor Sanh, and Léo Tronchon. Building and better understanding vision-language models: insights and future directions. *arXiv preprint arXiv:2408.12637*, 2024. 5

[8] Ji Lin, Hongxu Yin, Wei Ping, Yao Lu, Pavlo Molchanov, Andrew Tao, Huizi Mao, Jan Kautz, Mo-

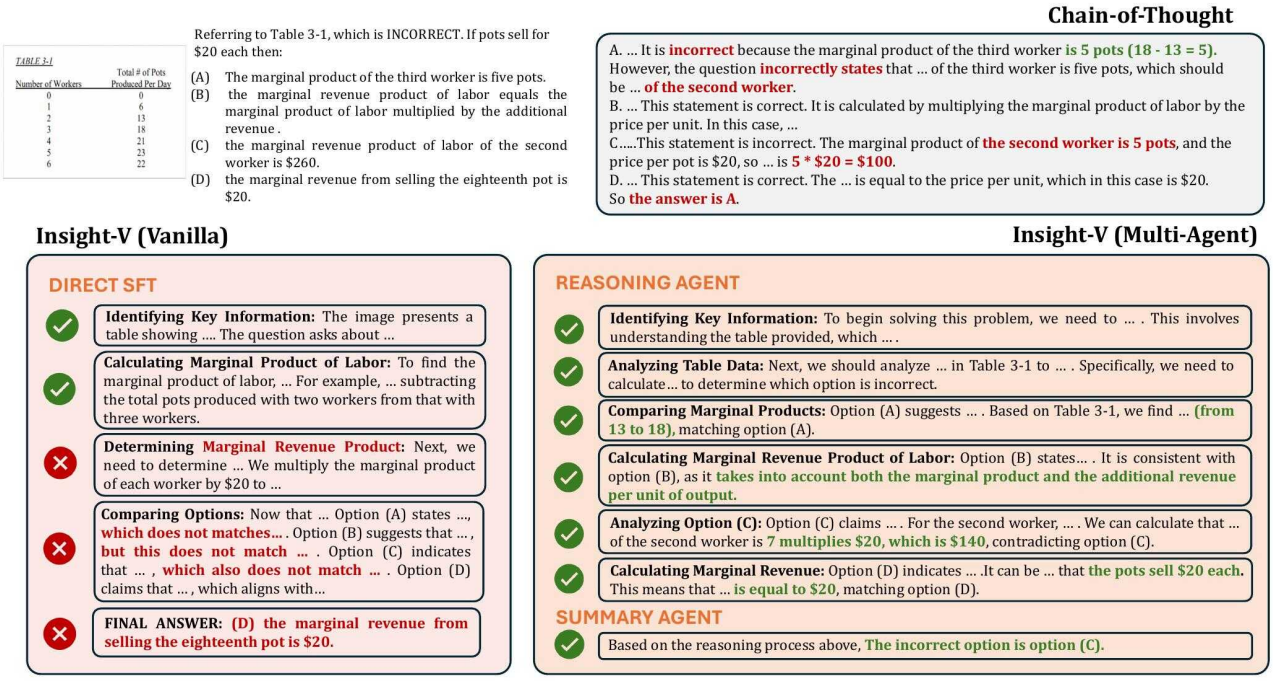

Figure 9. Additional qualitative cases spanning documents, charts, and arithmetic prompts. The reasoning agent explores alternatives while the summary agent overrides low-confidence paths, illustrating robustness across diverse layouts and tasks.

hammad Shoeybi, and Song Han. Vila: On pre-training for visual language models, 2023. 5

[9] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *CVPR*, pages 26296–26306, 2024. 1, 2, 3, 5

[10] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge, 2024. 3, 5

[11] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *NeurIPS*, 36, 2024. 1

[12] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, et al. Mmbench: Is your multi-modal model an all-around player? *arXiv preprint arXiv:2307.06281*, 2023. 2

[13] Yuliang Liu, Zhang Li, Biao Yang, Chunyuan Li, Xucheng Yin, Cheng-lin Liu, Lianwen Jin, and Xiang Bai. On the hidden mystery of ocr in large multimodal models. *arXiv preprint arXiv:2305.07895*, 2023. 2

[14] Yuan Liu, Zhongyin Zhao, Ziyuan Zhuang, Le Tian, Xiao Zhou, and Jie Zhou. Points: Improving your vision-language model with affordable strategies. *arXiv preprint arXiv:2409.04828*, 2024. 5

[15] Haoyu Lu, Wen Liu, Bo Zhang, Bingxuan Wang, Kai Dong, Bo Liu, Jingxiang Sun, Tongzheng Ren, Zhuoshu Li, Yaofeng Sun, et al. Deepseek-vl: towards real-world vision-language understanding. *arXiv preprint arXiv:2403.05525*, 2024. 5

[16] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. *arXiv preprint arXiv:2310.02255*, 2023. 2

[17] Shiyin Lu, Yang Li, Qing-Guo Chen, Zhao Xu, Weihua Luo, Kaifu Zhang, and Han-Jia Ye. Ovis: Structural embedding alignment for multimodal large language model. *arXiv:2405.20797*, 2024. 5

[18] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. *arXiv preprint arXiv:2203.10244*, 2022. 2

[19] Minesh Mathew, Dimosthenis Karatzas, and CV Jawahar. Docvqa: A dataset for vqa on document images. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 2200–2209, 2021. 2

[20] QwenTeam. Qwen2-vl: To see the world more clearly. *Wwen Blog*, 2024. 1, 3, 5

[21] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al.

Learning transferable visual models from natural language supervision. In *ICML*, pages 8748–8763. PMLR, 2021. 2, 3, 5

[22] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2024. 1, 2, 5

[23] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 8317–8326, 2019. 2

[24] Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, et al. Cambrian-1: A fully open, vision-centric exploration of multimodal llms. *arXiv preprint arXiv:2406.16860*, 2024. 5

[25] Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, Qianyu Chen, Huarong Zhou, Zhensheng Zou, Haoye Zhang, Shengding Hu, Zhi Zheng, Jie Zhou, Jie Cai, Xu Han, Guoyang Zeng, Dahai Li, Zhiyuan Liu, and Maosong Sun. Minicpm-v: A gpt-4v level mllm on your phone. *arXiv preprint 2408.01800*, 2024. 5

[26] Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, et al. Minicpm-v: A gpt-4v level mllm on your phone. *arXiv preprint arXiv:2408.01800*, 2024. 5

[27] Tianyu Yu, Haoye Zhang, Yuan Yao, Yunkai Dang, Da Chen, Xiaoman Lu, Ganqu Cui, Taiwen He, Zhiyuan Liu, Tat-Seng Chua, et al. Rlaif-v: Aligning mllms through open-source ai feedback for super gpt-4v trustworthiness. *arXiv preprint arXiv:2405.17220*, 2024. 5

[28] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9556–9567, 2024. 2

[29] Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Ming Yin, Botao Yu, Ge Zhang, et al. Mmmu-pro: A more robust multi-discipline multimodal understanding benchmark. *arXiv preprint arXiv:2409.02813*, 2024. 2

[30] Haotian Zhang, Mingfei Gao, Zhe Gan, Philipp Dufter, Nina Wenzel, Forrest Huang, Dhruti Shah, Xianzhi Du, Bowen Zhang, Yanghao Li, et al. Mm1. 5: Methods, analysis & insights from multimodal llm fine-tuning. *arXiv preprint arXiv:2409.20566*, 2024. 5

[31] Pan Zhang, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Rui Qian, Lin Chen, Qipeng Guo, Haodong Duan, Bin Wang, Linke Ouyang, Songyang Zhang, Wenwei Zhang, Yining Li, Yang Gao, Peng Sun, Xinyue Zhang, Wei Li, Jingwen Li, Wenhai Wang, Hang Yan, Conghui He, Xingcheng Zhang, Kai Chen, Jifeng Dai, Yu Qiao, Dahua Lin, and Jiaqi Wang. Internlm-xcomposer-2.5: A versatile large vision language model supporting long-contextual input and output. *arXiv preprint arXiv:2407.03320*, 2024. 5